

LAB: 07

This lab focuses on memory analysis.

Objective:

- Learn digital artifacts through Memory analysis with Volatility.

Learning Activities:

At the end of these activities, you should understand:

- Engage in standard processes of identifying, collecting, and analyzing relevant digital artifacts while keeping in mind the principles of sound forensic investigation, such as evidence preservation, documentation, evidence-based conclusions and artifacts.

Tools to use:

- Volatility (volatility_2.6_win64_standalone.zip -> <https://www.volatilityfoundation.org/26>)
- Magnet Forensics RAM Capture (<https://www.magnetforensics.com/resources/magnet-ram-capture/>)
- Python (<https://github.com/volatilityfoundation/volatility/blob/master/README.txt>)

Note: Please include a screenshot of each step. Crop your screenshot to show relevant information only. Uncropped screenshots will result in a 10% deduction from your marks.

As you know, attackers often target servers or exploit employee workstations or end-user devices through phishing, and these systems and devices are on practically all the time, or at least most of the day. As such, acquiring images and/or potential evidence from a live machine is a likely situation in many digital forensic investigations. However, when you encounter a victim system it may not be feasible to shut it down, or an intruder may still be poking around the system/network and you may not want to alert him or her to your presence. Besides, there is a plethora of potential evidence to be had from a victim system left running in its current state. The volatile contents of RAM should be captured as a snapshot of what is running on the system at that precise time, not to mention that it contains information that is not necessarily kept on the system's physical drive. Let's start here and capture the memory from the victim system.

There are many tools that can capture the memory from a live system, but we will be using FTK Imager Lite from AccessData. It is a GUI tool and compared to some other similarly purposed command-line tools, it leaves a larger footprint on the machine; however, all considering, its impact on the system is still rather minimal and it tends to collect more reliable images. As an investigator, you will have to decide if it is more acceptable to lose the volatile information contained in RAM or, alternatively, to interact with the system and alter some information on the physical drive.

Task 1:

Memory Capture

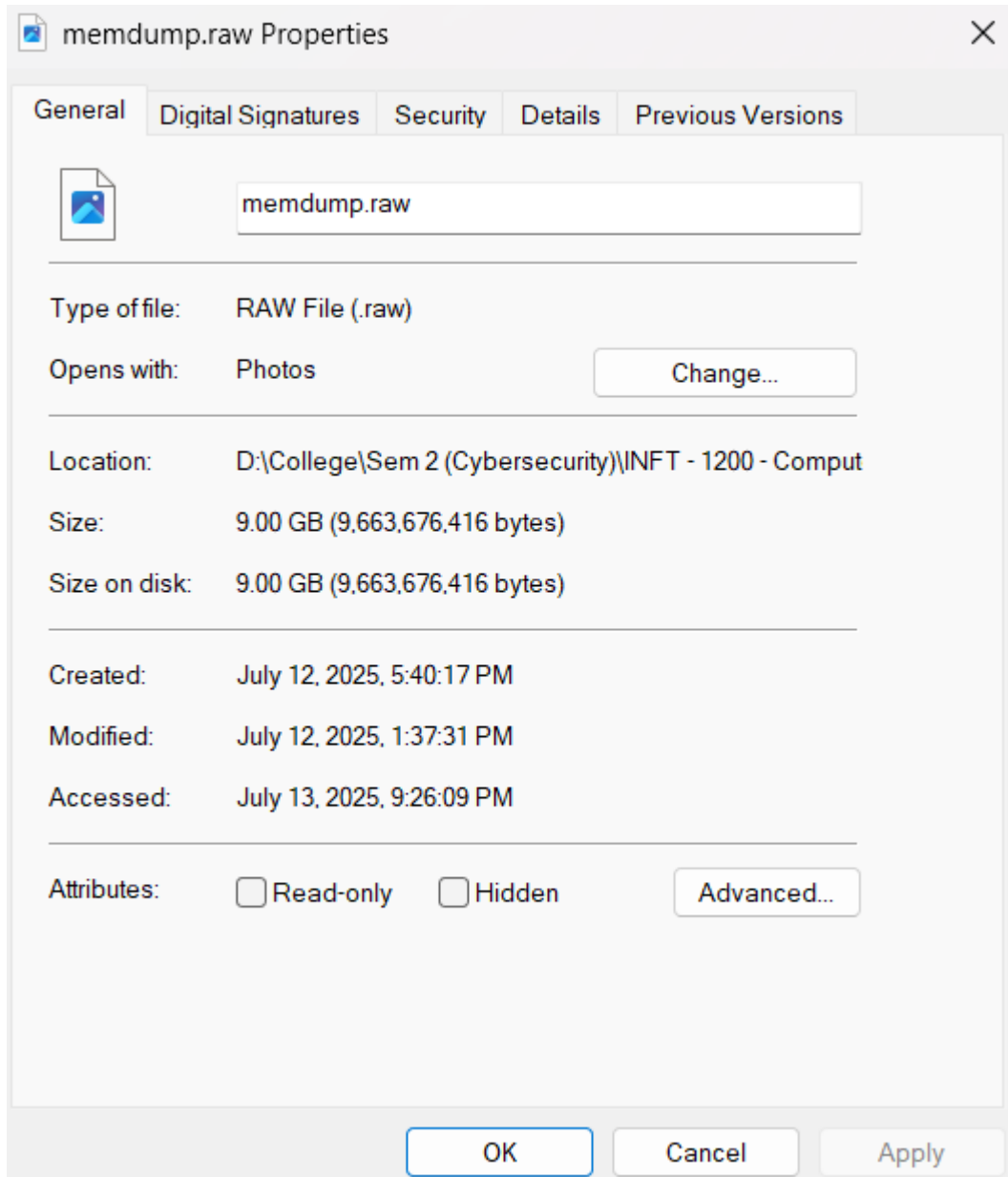
1.1 Insert your USB stick. If necessary, connect it to the VM. Navigate to it in Windows Explorer and execute FTK Imager Lite. Click Yes in the User Account Control window.

1.2. Find the little memory stick icon in the FTK Imager Lite toolbar. If you hold the mouse over it, it should say 'Capture Memory'. Click it to view the Memory Capture window.

1.3. Select Browse and set your Destination Path to somewhere on your removable USB drive. Then, you may change the name of the captured memory file to whatever you like. Note, FTK Imager will create a raw memory capture, just be aware of different formats used in different tools. Leave the boxes unchecked and click the Capture Memory button. This will take a little while as it is capturing XXGB of memory.

1.4 When the capture has completed, the Status line should read 'Memory capture finished successfully'. Now, you have captured the memory from a live system. Click Close, but keep FTK Imager Lite running. We will analyze the memory capture later.

What is the size of the file (captured mem file)? File Size is 24.9GB
(Take a screenshot of this result)



Note: You may use Magnet RAM Capture utility to capture memory.

Task-2:

Memory Analysis

For this part of the lab, you can disconnect your USB drive from the VM and make it accessible to your main operating system. We will be using Volatility Framework to analyze the raw memory dump that you saved to your USB drive. Volatility uses plugins, which makes it rather extensible, and the framework provides a powerful and dynamic memory analysis tool used by many investigators throughout the digital forensics community. I encourage you to explore this tool as much as possible. To

start, you might check out the basic usage instructions at the following link:
<https://www.forwarddefense.com/pdfs/Memory-Analysis-with-Volatility.pdf>.

2.1 Start a command shell and navigate to where your standalone Volatility executable is. Note, it may be helpful to have your memory image and the standalone Volatility exe in the same directory, whether on your USB drive or copied onto your main hard drive.

2.2 Typically, you would run the 'imageinfo' plugin to determine the proper profile to use with any particular memory image you may be analyzing, but since that can take some time. In the shell, enter the following command:

(Take a screenshot of this result)

```
Is64Bit True
IsPAE False
Layer_name 0 WindowsIntel32e
memory_layer 1 FileLayer
KdVersionBlock 0xf805d6c8a7c0
Major/Minor 15,26100
MachineType 34404
KeNumberProcessors 4
SystemTime 2025-07-12 17:36:13+00:00
NtSystemRoot C:\WINDOWS
NtProductType NtProductWinNt
NtMajorVersion 10
NtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine 34404
PE TimeDateStamp Wed Aug 29 17:06:46 2085
```

```
C:\support\volatility\volatility_2.6_win64_standalone>volatility_2.6_win64_standalone.exe imageinfo -f c:\support\memdump1.raw
```

Note: The above command may take up to 15 minutes to display the output.

Ex:

"C:\support\volatility\volatility_2.6_win64_standalone" is the volatility utility directory

"C:\support\memdump1.raw" is the captured mem file

```
C:\support\volatility\volatility_2.6_win64_standalone>volatility_2.6_win64_standalone.exe imageinfo -f c:\support\memdump1.raw
Volatility Foundation Volatility Framework 2.6
INFO : volatility.debug : Determining profile based on KDBG search...
Suggested Profile(s) : Win8SP0x64, Win81U1x64, Win2012R2x64_18340, Win2012R2x64, Win2012x64, Win8SP1x64_18340, Win8SP1x64 (Instantiated with Win8SP1x64)
AS Layer1 : WindowsAMD64PagedMemory (Kernel AS)
AS Layer2 : FileAddressSpace (C:\support\memdump1.raw)
PAE type : No PAE
DIB : 0x1aa000L
KDBG : 0xf800356c5530L
Number of Processors : 4
Image Type (Service Pack) : 0
KPCR for CPU 0 : 0xfffff80035722000L
KPCR for CPU 1 : 0xfffff8001e27bc000L
KPCR for CPU 2 : 0xfffff8001e6240000L
KPCR for CPU 3 : 0xfffff8001e25a3000L
KUSER_SHARED_DATA : 0xfffff78000000000L
Image date and time : 2021-03-14 04:32:03 UTC+0000
Image local date and time : 2021-03-13 23:32:03 -0500
C:\support\volatility\volatility_2.6_win64_standalone>
```

2.3 Next, enter the following command:

(Take a screenshot of this result)

```

C:\Forensics\Lab 7\Dmp 2\memdump.raw windows.netstat
Volatility 3 Framework 2.26.2
Progress: 100.00
Offset Proto LocalAddr LocalPort ForeignAddr ForeignPort State PID Owner Created
0x9e0f22738488 TCPv4 0.0.0.0 135 0.0.0.0 0 LISTENING 688 svchost.exe 2025-07-12 17:19:53.000000 UTC
0x9e0f22738488 TCPv6 :: 135 :: 0 LISTENING 688 svchost.exe 2025-07-12 17:19:53.000000 UTC
0x9e0f221bcc488 TCPv4 0.0.0.0 135 0.0.0.0 0 LISTENING 688 svchost.exe 2025-07-12 17:19:53.000000 UTC
0x9e0f22fe4248 TCPv4 192.168.91.131 139 0.0.0.0 0 LISTENING 4 System 2025-07-12 17:19:57.000000 UTC
0x9e0f23196be8 TCPv4 0.0.0.0 445 0.0.0.0 0 LISTENING 4 System 2025-07-12 17:20:01.000000 UTC
0x9e0f23196be8 TCPv6 :: 445 :: 0 LISTENING 4 System 2025-07-12 17:20:01.000000 UTC
0x9e0f2353d7c8 TCPv4 0.0.0.0 5840 0.0.0.0 0 LISTENING 1200 svchost.exe 2025-07-12 17:22:05.000000 UTC
0x9e0f23195748 TCPv4 0.0.0.0 7680 0.0.0.0 0 LISTENING 5932 svchost.exe 2025-07-12 17:23:34.000000 UTC
0x9e0f23195748 TCPv6 :: 7680 :: 0 LISTENING 5932 svchost.exe 2025-07-12 17:23:34.000000 UTC
0x9e0f21bcd4d8 TCPv4 0.0.0.0 49664 0.0.0.0 0 LISTENING 860 lsass.exe 2025-07-12 17:19:53.000000 UTC
0x9e0f21bcd4d8 TCPv6 :: 49664 :: 0 LISTENING 860 lsass.exe 2025-07-12 17:19:53.000000 UTC
0x9e0f21bcc068 TCPv4 0.0.0.0 49664 0.0.0.0 0 LISTENING 860 lsass.exe 2025-07-12 17:19:53.000000 UTC
0x9e0f22738e28 TCPv4 0.0.0.0 49665 0.0.0.0 0 LISTENING 696 wininit.exe 2025-07-12 17:19:53.000000 UTC
0x9e0f22738e28 TCPv6 :: 49665 :: 0 LISTENING 696 wininit.exe 2025-07-12 17:19:53.000000 UTC
0x9e0f227310e8 TCPv4 0.0.0.0 49665 0.0.0.0 0 LISTENING 696 wininit.exe 2025-07-12 17:19:53.000000 UTC
0x9e0f228c2328 TCPv4 0.0.0.0 49666 0.0.0.0 0 LISTENING 1536 svchost.exe 2025-07-12 17:19:55.000000 UTC
0x9e0f228c2328 TCPv6 :: 49666 :: 0 LISTENING 1536 svchost.exe 2025-07-12 17:19:55.000000 UTC
0x9e0f228c25e8 TCPv4 0.0.0.0 49666 0.0.0.0 0 LISTENING 1536 svchost.exe 2025-07-12 17:19:55.000000 UTC
0x9e0f228c3be8 TCPv4 0.0.0.0 49667 0.0.0.0 0 LISTENING 1900 svchost.exe 2025-07-12 17:19:55.000000 UTC
0x9e0f228c3be8 TCPv6 :: 49667 :: 0 LISTENING 1900 svchost.exe 2025-07-12 17:19:55.000000 UTC
0x9e0f228c3d48 TCPv4 0.0.0.0 49667 0.0.0.0 0 LISTENING 1900 svchost.exe 2025-07-12 17:19:55.000000 UTC
0x9e0f22fe3e28 TCPv4 0.0.0.0 49668 0.0.0.0 0 LISTENING 3064 spoolsv.exe 2025-07-12 17:19:58.000000 UTC
0x9e0f22fe3e28 TCPv6 :: 49668 :: 0 LISTENING 3064 spoolsv.exe 2025-07-12 17:19:58.000000 UTC
0x9e0f22fe4e48 TCPv4 0.0.0.0 49668 0.0.0.0 0 LISTENING 3064 spoolsv.exe 2025-07-12 17:19:58.000000 UTC
0x9e0f236ff7c8 TCPv4 0.0.0.0 49669 0.0.0.0 0 LISTENING 840 services.exe 2025-07-12 17:20:03.000000 UTC
0x9e0f236ff7c8 TCPv6 :: 49669 :: 0 LISTENING 840 services.exe 2025-07-12 17:20:03.000000 UTC
0x9e0f275558e8 UDPv4 0.0.0.0 123 * 0 2612 svchost.exe 2025-07-12 17:22:13.000000 UTC
0x9e0f275558e8 UDPv6 :: 123 * 0 2612 svchost.exe 2025-07-12 17:22:13.000000 UTC
0x9e0f27555a88 UDPv4 0.0.0.0 123 * 0 2612 svchost.exe 2025-07-12 17:22:13.000000 UTC
0x9e0f22ea9798 UDPv4 192.168.91.131 137 * 0 4 System 2025-07-12 17:19:57.000000 UTC
0x9e0f22ea7a58 UDPv4 192.168.91.131 138 * 0 4 System 2025-07-12 17:19:57.000000 UTC
0x9e0f242cd6e8 UDPv6 fe80::d9dd:211a:56ef:922b 1900 * 0 1812 svchost.exe 2025-07-12 17:20:26.000000 UTC
0x9e0f242cb9a8 UDPv6 :: 1 1900 * 0 1812 svchost.exe 2025-07-12 17:20:26.000000 UTC

```

C:\support\volatility\volatility_2.6_win64_standalone>volatility_2.6_win64_standalone.exe --profile=Win2012R2x64_18340 -f c:\support\memdump1.raw netscan

Note, it is often helpful to send the output to a text file, especially if you are using a Windows command shell, it's just easier to read. The above command, which uses the 'netstat' plugin, lists all network connections, protocols, IP addresses, ports, and associated processes.

What is the command to send the output of the above command to a text file for later analysis?

For current scenario: "python vol.py -f "D:\College\Sem 2 (Cybersecurity)\INFT - 1200 - Computer Forensics\LAB 7\Dmp 2\memdump.raw" windows.netstat > netstat_output.txt"

However, general command would be "python vol.py -f "path" windows.netstat > filename.txt", to put it simple most of the outputs can be save to a file using ">" followed by "filename.extension" in python although some of the software may not be able to send the output while executing command.

2.4 To locate the virtual addresses of registry hives in memory, and the full paths to the corresponding hive on disk, use the hivelist command. If you want to print values from a certain hive, run this command first so you can see the address of the hives.

(Take a screenshot of this result)

```

.dat Disabled
0xc50bbac46000 \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\MicrosoftWindows.Client.00BE_1000.26100.2.0_x64__cw5n1h2txyewy\ActivationStore.dat Disabled
0xc50bbac49000 \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\MicrosoftWindows.Client.FileExp_1000.26100.3.0_x64__cw5n1h2txyewy\ActivationStore.dat Disabled
0xc50bbac4c000 \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\MicrosoftWindows.Client.Core_1000.26100.19.0_x64__cw5n1h2txyewy\ActivationStore.dat Disabled
0xc50bbac50000 \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\MicrosoftWindows.Client.CBS_1000.26100.18.0_x64__cw5n1h2txyewy\ActivationStore.dat Disabled
0xc50bbade9000 \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.UI.Xaml.CBS_9.2311.10002.0_x64__8wekyb3d8bwe\ActivationStore.dat Disabled
0xc50bbbaecc000 \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\Microsoft.Windows.StartMenuExperienceHost_10.0.26100.1301_neutral_neutral_cw5n1h2txyewy\ActivationStore.dat Disabled
0xc50bbbaece000 \??\C:\Users\Night\AppData\Local\Packages\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\Settings\settings.dat Disabled
0xc50bbbf36000 \??\C:\ProgramData\Microsoft\Windows\AppRepository\Packages\MicrosoftWindows.Client.WebExperience_525.10401.30.0_x64__cw5n1h2txyewy\ActivationStore.dat Disabled
0xc50bbfafa000 \??\C:\ProgramData\Packages\MicrosoftWindows.Client.WebExperience_cw5n1h2txyewy\S-1-5-21-1279783837-385767984-3735737057-1001\SystemAppData\Helium\Cache\2949f22fc3f6f7e.dat Disabled
0xc50bbfbfb000 \??\C:\Users\Night\AppData\Local\Packages\MicrosoftWindows.Client.WebExperience_cw5n1h2txyewy\SystemAppData\Helium\User.dat Disabled

```

C:\support\volatility\volatility_2.6_win64_standalone>volatility_2.6_win64_standalone.exe --profile=Win2012R2x64_18340 -f c:\support\memdump1.raw hivelist

```
C:\support\volatility\volatility_2.6_win64_standalone>volatility_2.6_win64_standalone.exe --profile=Win2012R2x64_18340 -f c:\support\memdump1.raw hivelist
Volatility Foundation Volatility Framework 2.6
-----
Virtual      Physical      Name
-----
0xfffff0001561c000 0x0000000000378000 [no name]
0xfffff00015631000 0x000000000018c000 \REGISTRY\MACHINE\SYSTEM
0xfffff0001564b000 0x000000000021a000 \REGISTRY\MACHINE\HARDWARE
0xfffff00016a89000 0x000000013956a000 \SystemRoot\System32\Config\SAM
0xfffff00016a7f000 0x000000012e210000 \SystemRoot\System32\Config\DEFAULT
0xfffff00016a8e000 0x000000013926d000 \SystemRoot\System32\Config\SECURITY
0xfffff00016a7d000 0x00000001384bd000 \SystemRoot\System32\Config\SOFTWARE
0xfffff0001aa8e000 0x000000010475f000 \Device\HarddiskVolume1\Boot\BCD
0xfffff00017c33000 0x0000000136239000 ??\C:\Windows\ServiceProfiles\NetworkService\NTUSER.DAT
0xfffff00017c7000 0x0000000000234000 \SystemRoot\System32\Config\BBT
0xfffff00017c26000 0x00000001016e9000 ??\C:\Windows\ServiceProfiles\LocalService\NTUSER.DAT
0xfffff00018aa2000 0x000000012cc12000 ??\C:\Users\Administrator\AppData\Local\Microsoft\Windows\UsrClass.dat
0xfffff00018c0f000 0x000000012a562000 ??\C:\Users\Administrator\AppData\Local\Microsoft\Windows\UsrClass.dat
0xfffff00018ce000 0x000000012d3ea000 ??\C:\Users\Administrator\ntuser.dat
0xfffff00018d45000 0x000000012dd0f000 ??\C:\Users\Administrator\AppData\Local\Microsoft\Windows\UsrClass.dat
0xfffff0001940000 0x00000001306ab000 ??\C:\Windows\AppData\Programs\hmcache.hve
C:\support\volatility\volatility_2.6_win64_standalone>
```

2.5 The next command would output list multiple hives (DEFAULT and ntuser.dat) contain the same key "Software\Microsoft\Windows NT\CurrentVersion". (Take a screenshot of this result)

```
0xc50bae2f0000 Key [NONAME]\Software\Microsoft\Windows NT\CurrentVersion - - -
0xc50bae2ed000 Key \REGISTRY\MACHINE\SYSTEM\Software\Microsoft\Windows NT\CurrentVersion - - -
0xc50bae370000 Key \REGISTRY\MACHINE\HARDWARE\Software\Microsoft\Windows NT\CurrentVersion - - -
0xc50bb07b5000 Key \SystemRoot\System32\Config\SOFTWARE\Software\Microsoft\Windows NT\CurrentVersion - - -
0xc50bb0a6b000 Key \Device\HarddiskVolume1\EFI\Microsoft\Boot\BCD\Software\Microsoft\Windows NT\CurrentVersion - - -
0xc50bb0777000 Key \SystemRoot\System32\Config\DEFAULT\Software\Microsoft\Windows NT\CurrentVersion - - -
0xc50bb07b3000 Key \SystemRoot\System32\Config\SECURITY\Software\Microsoft\Windows NT\CurrentVersion - - -
0xc50bb0775000 Key \SystemRoot\System32\Config\SAM\Software\Microsoft\Windows NT\CurrentVersion - - -
0xc50bb22b4000 Key ??\C:\WINDOWS\ServiceProfiles\NetworkService\NTUSER.DAT\Software\Microsoft\Windows NT\CurrentVersion - - -
0xc50bb2620000 Key \SystemRoot\System32\Config\BBT\Software\Microsoft\Windows NT\CurrentVersion - - -
```

C:\support\volatility\volatility_2.6_win64_standalone>volatility_2.6_win64_standalone.exe --profile=Win2012R2x64_18340 -f c:\support\memdump1.raw printkey -v -K "Software\Microsoft\Windows NT\CurrentVersion"

2.6 Next, the following command would display a list of running process:

C:\support\volatility\volatility_2.6_win64_standalone>volatility_2.6_win64_standalone.exe --profile=Win2012R2x64_18340 -f c:\support\memdump1.raw pslist

(Take a screenshot of this result)

PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	File output
4	0	System	0x9e0f1f6b1040	186	-	N/A	False	2025-07-12 17:19:41.000000 UTC	N/A	Disabled
124	4	Registry	0x9e0f1f740080	4	-	N/A	False	2025-07-12 17:19:32.000000 UTC	N/A	Disabled
436	4	smss.exe	0x9e0f21c0e040	2	-	N/A	False	2025-07-12 17:19:41.000000 UTC	N/A	Disabled
624	596	csrss.exe	0x9e0f221bd080	11	-	0	False	2025-07-12 17:19:50.000000 UTC	N/A	Disabled
696	596	wininit.exe	0x9e0f224a3100	2	-	0	False	2025-07-12 17:19:51.000000 UTC	N/A	Disabled
784	688	csrss.exe	0x9e0f22465080	13	-	1	False	2025-07-12 17:19:51.000000 UTC	N/A	Disabled
768	688	winlogon.exe	0x9e0f22510080	4	-	1	False	2025-07-12 17:19:51.000000 UTC	N/A	Disabled
840	696	services.exe	0x9e0f2253d080	8	-	0	False	2025-07-12 17:19:52.000000 UTC	N/A	Disabled
860	696	lsass.exe	0x9e0f2254d080	11	-	0	False	2025-07-12 17:19:52.000000 UTC	N/A	Disabled
988	840	svchost.exe	0x9e0f2261e240	16	-	0	False	2025-07-12 17:19:53.000000 UTC	N/A	Disabled
544	768	fontdrvhost.exe	0x9e0f22636080	5	-	1	False	2025-07-12 17:19:53.000000 UTC	N/A	Disabled
492	696	fontdrvhost.exe	0x9e0f22633080	5	-	0	False	2025-07-12 17:19:53.000000 UTC	N/A	Disabled
688	840	svchost.exe	0x9e0f227020c0	13	-	0	False	2025-07-12 17:19:53.000000 UTC	N/A	Disabled
692	840	svchost.exe	0x9e0f2274f080	6	-	0	False	2025-07-12 17:19:53.000000 UTC	N/A	Disabled
1104	768	dmw.exe	0x9e0f22664080	17	-	1	False	2025-07-12 17:19:54.000000 UTC	N/A	Disabled
1188	840	svchost.exe	0x9e0f228020c0	11	-	0	False	2025-07-12 17:19:54.000000 UTC	N/A	Disabled

2.7 Next command would display the process tree:

C:\support\volatility\volatility_2.6_win64_standalone>volatility_2.6_win64_standalone.exe --
profile=Win2012R2x64_18340 -f c:\support\memdump1.raw pstree

(Take a screenshot of this result)

PID	PPID	ImageFileName	Offset(V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	Audit	Cmd	Pat
4	0	System	0x9e0f1f6b1040	186	-	N/A	False	2025-07-12 17:19:41.000000 UTC	N/A	-	-	-
* 436	4	smss.exe	0x9e0f21c0e040	2	-	N/A	False	2025-07-12 17:19:41.000000 UTC	N/A	-	-	\Device\HarddiskVol
em32\smss.exe		\SystemRoot\System32\smss.exe				\SystemRoot\System32\smss.exe						
* 124	4	Registry	0x9e0f1f740080	4	-	N/A	False	2025-07-12 17:19:32.000000 UTC	N/A	Registry	-	-
* 2132	4	MemCompression	0x9e0f22d19040	46	-	N/A	False	2025-07-12 17:19:55.000000 UTC	N/A	MemCompression	-	-
624	596	csrss.exe	0x9e0f221bd080	11	-	0	False	2025-07-12 17:19:50.000000 UTC	N/A	-	-	\Device\HarddiskVol
em32\csrss.exe		%SystemRoot%\system32\csrss.exe				ObjectDirectory=\Windows SharedSection=1024,20480,768 Windows=On SubSystemType=Windows Serv						
erverDll=winsrv:UserServerDllInitialization,3 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16												
696	596	wininit.exe	0x9e0f224a3140	2	-	0	False	2025-07-12 17:19:51.000000 UTC	N/A	-	-	\Device\HarddiskVol
em32\wininit.exe		C:\WINDOWS\system32\wininit.exe										
* 840	696	services.exe	0x9e0f2253d080	8	-	0	False	2025-07-12 17:19:52.000000 UTC	N/A	-	-	\Device\HarddiskVol
em32\services.exe		C:\WINDOWS\system32\services.exe				C:\WINDOWS\system32\services.exe						
** 1536	840	svchost.exe	0x9e0f2297d080	12	-	0	False	2025-07-12 17:19:54.000000 UTC	N/A	-	-	\Device\HarddiskVol
em32\svchost.exe		C:\WINDOWS\system32\svchost.exe				-k netsvcs -p -s Schedule						
*** 1780	1536	taskhostw.exe	0x9e0f24419080	6	-	0	False	2025-07-12 17:20:26.000000 UTC	N/A	-	-	\Device\Har
ows\System32\taskhostw.exe		taskhostw.exe				C:\WINDOWS\system32\taskhostw.exe						
*** 1588	1536	taskhostw.exe	0x9e0f24f4e080	7	-	1	False	2025-07-12 17:22:27.000000 UTC	N/A	-	-	\Device\Har

2.8 To view the SIDs associated with a process, use the getsids command:

C:\support\volatility\volatility_2.6_win64_standalone>volatility_2.6_win64_standalone.exe --
profile=Win2012R2x64_18340 -f c:\support\memdump1.raw getsids

(Take a screenshot of this result)

8652	cmd.exe	S-1-5-21-1279783837-385767984-3735737057-1001	Might
8652	cmd.exe	S-1-16-12288	High Mandatory Level
8652	cmd.exe	S-1-1-0	Everyone
8652	cmd.exe	S-1-5-114	Local Account (Member of Administrators)
8652	cmd.exe	S-1-5-32-544	Administrators
8652	cmd.exe	S-1-5-32-545	Users
8652	cmd.exe	S-1-5-4	Interactive
8652	cmd.exe	S-1-2-1	Console Logon (Users who are logged onto the physical console)
8652	cmd.exe	S-1-5-11	Authenticated Users
8652	cmd.exe	S-1-5-15	This Organization
8652	cmd.exe	S-1-11-96-3623454863-58364-18864-2661722203-1597581903-385978653-3976556724-2977520931-272651202-519333130	-
8652	cmd.exe	S-1-5-113	Local Account
8652	cmd.exe	S-1-5-5-0-959186	Logon Session
8652	cmd.exe	S-1-2-0	Local (Users with the ability to log in locally)
8652	cmd.exe	S-1-5-64-36	-
2780	SrTasks.exe	S-1-5-18	Local System
2780	SrTasks.exe	S-1-16-16384	System Mandatory Level
2780	SrTasks.exe	S-1-1-0	Everyone
2780	SrTasks.exe	S-1-5-32-545	Users

2.9 What is the command to display the loaded DLLs? python vol.py -f "path to dump file" windows.dlllist

3.0 What is the command to display process privileges detail? python vol.py -f "path to dump file" windows.privileges

Reflective statements (end-of-exercise):

You should reflect on these questions:

1. How can the Forensics Investigators preserve volatile evidence at the crime scene?

ANS: To save and retrieve volatile data on multiple drives, forensic examiners rely on a variety of equipment such as FTK imager. Time and process logs, connection data, and other crucial data that can be useful as evidence are some of the critical data that is saved in this way.

2. How can the Forensics Investigators make use of Artificial Intelligence (AI) and Machine Learning (ML) in digital forensics?

ANS: The vast amount of data received from computers makes it easy to sift through and analyze the data using AI and ML, significantly decreasing the amount of time spent gathering evidence and easier for forensic examiners to determine what to look for within the data, where to look for additional evidence, and what evidence to give.

3. What is the difference between artifacts and evidence?

ANS: Evidence is data or information that is relevant to the case or that is admissible to be heard in court, whereas artefacts are leftovers or bits of data that remain in the memory or storage of the recovered device and is the subject of a forensic investigation.

References

1. The Volatility Foundation. (n.d.). Volatility 2.6 Documentation.
<https://www.volatilityfoundation.org/26>
2. Wikipedia contributors. (2024, June 10). Memory forensics. Wikipedia.
https://en.wikipedia.org/wiki/Memory_forensics
3. SANS Institute. (2006). Techniques and tools for recovering and analyzing data from volatile memory. <https://www.sans.org/reading-room/whitepapers/forensics/techniques-tools-recovering-analyzing-data-volatile-memory-33049>
4. ScienceDirect. (n.d.). Memory forensics. In Computer Science Topics.
<https://www.sciencedirect.com/topics/computer-science/memory-forensics>